

Predictability Before Prediction: A Selective-Prediction Framework for Household SKU Choice

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Abstract

Most household product-choice models are evaluated as if a prediction should be issued for every eligible purchase event. This working paper studies a different operational question: **can pre-event purchase history identify which household × product events are predictable enough to answer at all?** The task is deliberately narrow.

Conditional on a household making a purchase within a fixed six-SKU choice set, the target is the SKU selected on that occasion; the study does not model purchase incidence or the composition of a full next basket.

Three predictors were developed in a nine-product panel. A market-prior baseline (B0) achieved 37.28% accuracy, a household-history model (B1) 64.53%, and a behavior-fusion model (B2) 65.52% across 17,408 forward-test events. In a separate 26,659-event practical-complexity study, B2 exceeded B1 by 1.2604 percentage points, with positive household- and week-cluster bootstrap intervals, but remained below a predeclared 3.0-point material-gain threshold. The public system therefore retains B1 as the simple predictor.

The study then freezes P1, a parameter-free pre-event household-stability score based on history maturity, choice concentration, entropy, switching, and recent concentration, together with three operating cutoffs. In a first new-product batch, the frozen rule achieved 39.16% coverage at 76.18% accuracy, 21.49% at 83.55%, and 11.50% at 88.14%. In a second untouched representative batch containing 40,474 events from 99 evaluable product groups, the corresponding results were 41.08%/77.17%, 22.00%/84.78%, and 11.01%/90.06%. A separate department challenge reached 79.33% accuracy at 40.22% coverage, versus 61.18% for B1 on all events. The evidence supports a simple selective-prediction architecture: use pre-event household stability to decide when to make a conditional SKU-choice prediction and **ABSTAIN** otherwise. The current evidence grade is

CONFIRMED_WITHIN_PANEL_CROSS_PRODUCT; independent retailer replication remains the next test.

The result also defines a staged research and application agenda: external replication first, followed by tests of purchase incidence, promotion and price response, decision value, and only then broader commercialization.

Keywords: selective prediction; reject option; consumer choice; household purchase history; SKU choice; retail analytics; abstention; reproducibility

1. Introduction

Consumer choice has long been modeled as a discrete decision among alternatives, with panel and scanner data providing unusually detailed evidence about repeated household purchases. Classical work showed that household-specific loyalty, marketing variables, and state dependence can explain substantial persistence in brand choice (Guadagni and Little, 1983; Erdem and Keane, 1996; Keane, 1997). Later work emphasized that many economically relevant decisions occur at the stock-keeping-unit (SKU) level rather than only at the brand level (Fader and Hardie, 1996), and that repeated choices can reflect both stable heterogeneity and dynamic state dependence across categories (Seetharaman, Ainslie, and Chintagunta, 1999).

The usual modeling question is: **given an event, which model predicts the choice most accurately?** That framing implicitly treats every eligible event as something the system should answer. In many deployed systems, however, a second decision is equally important: **should the system answer this event at all?** A prediction made on an event with weak historical signal can be less useful than an explicit abstention, especially when downstream users care about reliability as well as coverage.

This second decision connects household choice modeling to the literature on classification with a reject option or selective prediction. In that literature, a predictor is permitted to abstain on some observations so that performance can improve on the retained subset, producing a risk-coverage trade-off rather than a single all-case accuracy number (Chow, 1970; El-Yaniv and Wiener, 2010). Modern selective-prediction work develops the same idea for learned classifiers and deep networks (Cortes, DeSalvo, and Mohri, 2016; Geifman and El-Yaniv, 2017, 2019).

The present study applies that operational logic to repeated household SKU choice. It does **not** claim to introduce a new discrete-choice estimator, nor does it attempt to identify a causal mechanism of loyalty, inertia, or preference. Instead, it asks whether simple historical structure can be used prospectively to rank the reliability of a simple conditional-choice predictor.

The empirical strategy has three stages. First, a market-prior model (B0), a household-history model (B1), and a more complex behavior-fusion model (B2) are compared under forward evaluation. Second, a practical-complexity rule determines whether B2's additional accuracy is large enough to justify making it the default. Third, a separate pre-event score, P1, is frozen and used as an abstention gate. P1 does not change the predicted class; it determines whether B1 is allowed to answer.

This structure produces three main findings. First, household history provides the dominant predictive increment over a market prior in the development panel: B1 improves accuracy from 37.28% to 64.53%. Second, additional recent-behavior modeling improves average accuracy only modestly under the frozen practical rule, so the simpler B1 remains the default. Third, predictability is strongly heterogeneous across events. A fixed P1 gate reproduces a stable coverage-accuracy frontier across two non-overlapping new-product cohorts and a cross-department challenge.

The strongest result is not a claim that all household product choice is highly predictable. It is the narrower claim that **a subset of conditional purchase events can be identified before the outcome as materially more predictable than the full event population.** At the 0.6 operating point, for example, the untouched confirmatory batch accepts 41.08% of evaluable events and reaches 77.17% accuracy, compared with 59.37% for B1 when it answers all events in that cohort.

The rest of the paper positions this result against prior work, defines the task and frozen evaluation boundary, describes B0/B1/B2 and P1, reports the two generalization cohorts and department challenge, and documents negative results and limitations. Because the current evidence comes from new products inside the same panel ecosystem, the final section defines an external zero-shot replication protocol rather than treating within-panel confirmation as cross-retailer validation.

2. Related literature and research position

2.1 Discrete choice and repeated household purchase behavior

Discrete-choice models provide the basic framework for analyzing choices among a finite set of alternatives (McFadden, 1974). In scanner-panel marketing research, Guadagni and Little (1983) demonstrated that parsimonious multinomial choice models could combine household loyalty measures with price and promotion variables to explain brand-size choice. Subsequent work developed richer accounts of learning, heterogeneity, and state dependence in repeated consumer choice. Erdem and Keane (1996) modeled dynamic brand choice under uncertainty, while Keane (1997) explicitly separated persistent preference heterogeneity from true state dependence. Seetharaman, Ainslie, and Chintagunta (1999) examined household state-dependence patterns across categories, and Chintagunta (1999) integrated variety seeking and purchase timing into dynamic brand-choice analysis.

This literature establishes an important interpretive boundary for the present work. Historical household choice is predictively informative, but repeated choice can arise from multiple mechanisms. A strong history-based predictor does not by itself prove causal loyalty, inertia, habit formation, or low responsiveness to price and promotion. The present paper therefore treats household history as a predictive state representation and avoids causal interpretation.

Fader and Hardie (1996) further emphasized that the SKU, rather than the brand alone, can be the relevant unit of consumer choice. That perspective is directly aligned with the present task, which predicts one selected SKU from a fixed six-SKU set. At the same time, the choice sets here are empirically constructed substitute proxies; they should not be interpreted as a structural model of every item available on the shelf at the moment of purchase.

2.2 Consumer heterogeneity and simple personalization

Marketing models routinely recognize substantial heterogeneity across consumers (Allenby and Rossi, 1998). The empirical question here is intentionally narrower than estimating a full heterogeneous structural model. B1 asks whether a household's own pre-event counts, smoothed toward an external market prior, capture enough stable regularity to improve prediction. This is a deliberately low-complexity personalization strategy. The design is motivated by the practical possibility that, for many recurring purchase contexts, a simple household history may capture most of the usable predictive signal even when richer behavioral models remain statistically informative.

The B1-versus-B2 comparison is therefore not a contest between “simple” and “correct” consumer theory. B2 adds recent-choice features that can represent short-run behavioral dynamics, but the publication rule asks a deployment question: is the average incremental improvement large enough to justify the additional complexity? The threshold is operational rather than a universal statistical principle.

2.3 Selective prediction and abstention

The second literature is classification with rejection. Chow (1970) formalized the trade-off between recognition error and rejection. El-Yaniv and Wiener (2010) later developed a risk-coverage framework in which a selective classifier chooses both a prediction function and a selection function. Modern work extends the idea to learned rejection functions and deep networks (Cortes, DeSalvo, and Mohri, 2016; Geifman and El-Yaniv, 2017, 2019).

MRSP adopts the operational core of this framework: the system may abstain. In this paper, coverage is the fraction of **valuable household × product purchase events** for which the gate allows B1 to answer; selective accuracy is accuracy only on accepted events. Equivalently, one could express the same frontier as coverage versus selective error (one minus accuracy). The gate is not trained jointly with the predictor, and P1 is not a neural confidence score. It is a fixed, interpretable history-stability score. The contribution is therefore empirical and architectural: applying a frozen reject-option logic to conditional household SKU choice and testing whether the resulting frontier transfers across non-overlapping product cohorts.

2.4 Position of this study

The study sits between predictive consumer analytics and selective prediction. It does not attempt to outperform the broader literature on structural choice modeling, promotion response, next-basket prediction, or modern recommender systems. Instead, it tests a specific proposition: **before an outcome is observed, can simple household history identify conditional SKU-choice events on which a simple predictor is substantially more reliable?**

The contribution should be read as a reproducible empirical finding with a deliberately bounded claim. The paper's strongest validation is within-panel, cross-product confirmation. A genuinely independent retailer or consumer panel is required before claims about cross-environment transport are warranted.

3. Task, data, and evaluation design

3.1 Prediction task

Each evaluable observation is a purchase event associated with a household and a predefined six-SKU product group. Conditional on the fact that one of these six SKUs was purchased, the target is the selected SKU index. The task therefore differs from three related problems:

1. **purchase incidence** — whether the household will purchase the category or group at all;
2. **next-basket prediction** — which collection of products will appear on a future shopping trip; and
3. **open-set recommendation** — ranking products from a large or changing catalog.

None of those broader tasks is claimed in this release.

3.2 Data environment

The source environment is the 84.51° Complete Journey data ecosystem, distributed in the `completejourney` R package and related source files (Boehmke and Mortimer, 2025). The upstream transaction source used in the research pipeline contains 1,469,307 transaction line items across 2,469 households over 53 weeks, together with product metadata and promotion files. The present analysis does not use every transaction as a labeled choice event. It constructs product groups and eligible household $\times$ product purchase events under frozen rules.

Raw transaction data are not redistributed with this working paper because the exact redistribution rights of the local source files have not been closed at the per-file provenance level. The public package instead contains code, configuration, hashes, aggregate results, synthetic fixtures, and an external replication runner. This restriction affects redistribution, not the internal reproducibility audit of the reported numbers.

3.3 Six-SKU product groups

Product groups are constructed as six-SKU substitute proxies using historical product metadata and support. The construction is intentionally empirical. It creates a consistent six-alternative prediction task, but it does not claim that the six SKUs exhaust the household’s true real-time choice set or encode full shelf availability.

The development stage used nine product groups. Later evaluation used non-overlapping new-product cohorts drawn from a pre-outcome candidate universe. Candidate selection for the new-product batches uses information available through week 22; later W3-W6 outcomes are evaluation-only. Stable keys prevent product groups from being reused across the two main generalization batches.

3.4 Walk-forward evaluation and frozen boundaries

Evaluation uses six walk-forward blocks over the 53-week period. Each block has a FIT segment, a subsequent TUNE segment, and a later EVAL segment. The public new-product evidence reports W3-W6 evaluation blocks. The design preserves chronological order: model state for an event is reconstructed only from information available before that event.

The second representative batch is the central confirmatory test. Before it is evaluated, the following elements are frozen:

- the B0 market-prior definition;
- the B1 household-history model and smoothing grid;
- the P1 score formula;
- the three P1 cutoffs;
- the time windows;
- the product selection procedure; and

- the primary success and breadth rules.

B2, learned gates, and P1 recalibration are excluded from the confirmatory run. This separation is important because the project involved extensive development work before the frozen confirmation stage.

4. Models and selective-prediction rule

4.1 B0: market prior

For household h , event t , and alternative j , B0 forms a smoothed market prior from historical global counts while excluding the current household's own history:

$$\text{other_count_j} = \max(\text{global_count_j_before_t} - \text{household_count_hj_before_t}, 0) + 1$$

$$\text{B0_j} = \text{other_count_j} / \text{sum_l}(\text{other_count_l})$$

The +1 term provides smoothing. The predicted SKU is the alternative with the largest B0 probability.

4.2 B1: household-history predictor

B1 combines the external market prior with the household's own prior choices:

$$\mathbf{x_j} = \text{lambda_bin} * \text{B0_j} + \text{household_count_hj_before_t}$$

$$\text{B1_j} = \mathbf{x_j} / \text{sum_l}(\mathbf{x_l})$$

The smoothing parameter `lambda_bin` is selected from a fixed grid using only past FIT/TUNE data and is indexed by three history-maturity bins: 0-2, 3-9, and 10+ prior purchases. No current-event label enters the construction.

B1 is not intended as a structural model of preference formation. It is a deliberately simple probabilistic history model whose main role is to establish how much predictive value can be obtained from pre-event household-specific choice history.

4.3 B2: behavior-fusion model

B2 begins with B1 and adds recent-behavior information such as the last choice, repeat streak, recent choice shares, and elapsed time since prior choices. The model therefore tests whether short-run behavioral dynamics produce enough incremental predictive value to justify extra complexity.

Across 26,659 nonzero-history events, B1 accuracy is 64.4210% and B2 accuracy is 65.6814%, a difference of 1.2604 percentage points. Household-cluster and week-cluster bootstrap 95% intervals are both positive (0.8568 to 1.6575 points and 0.7541 to 1.7492 points, respectively). The project nevertheless retains B1 because the gain does not clear the predeclared 3.0-point practical-complexity threshold.

This decision should not be interpreted as evidence that B2 contains no information. It states that, under the frozen deployment rule, the average gain is not large enough to replace B1 as the default public predictor.

4.4 P1: pre-event household-stability score

Let k be the number of prior purchases by the household in the current six-SKU group. Define

$$\text{maturity} = \text{clip}(\log(1+k) / \log(21), 0, 1)$$

$$\begin{aligned} \text{core} = & 0.30 * \text{top_share} \\ & + 0.25 * (1 - \text{normalized_entropy}) \\ & + 0.25 * (1 - \text{switch_rate}) \end{aligned}$$

$$+ 0.20 \cdot \text{recent_concentration}$$

$$P1 = \text{clip}(\text{maturity} \cdot \text{core}, 0, 1)$$

`top_share` is the largest historical share among the six candidates. `normalized_entropy` is household choice entropy divided by $\log(6)$. `switch_rate` is the proportion of adjacent historical choices that differ.

`recent_concentration` is the largest candidate share among the five most recent purchases. Every component uses only information available before the current event. P1 has no learned coefficients and no product-specific fitted weights.

Three cutoffs were calibrated on an older nine-product historical segment and then frozen:

Operating label	Frozen P1 cutoff	Historical coverage	Historical accuracy
0.60	0.3608488067	64.07%	69.06%
0.70	0.5231119438	40.00%	76.96%
0.80	0.6660270792	20.01%	86.93%

The labels 0.60, 0.70, and 0.80 are names for calibration operating points; they are not guarantees of future accuracy.

4.5 Selective decision rule

For a frozen cutoff c :

- if P1 is at least c , output the B1 prediction;
- if P1 is below c , output **ABSTAIN**.

Coverage is the accepted-event fraction. Accuracy is calculated only on accepted events. P1 therefore ranks event-level predictability; it does not redefine the predicted class.

5. Results

5.1 Development baselines

Table 1 shows the original nine-product development comparison.

Model	Forward-test events	Accuracy
B0 — market prior	17,408	37.28%
B1 — household history	17,408	64.53%
B2 — behavior fusion	17,408	65.52%

The principal development result is the B1-B0 difference of approximately 27.25 percentage points. This establishes that household-specific history contains substantial predictive information beyond a pooled market prior in this task. Because the analysis is predictive, this result is consistent with — but does not distinguish among — persistent preference heterogeneity, state dependence, habit, or other mechanisms discussed in the consumer-choice literature.

5.2 First new-product generalization batch

The first generalization cohort selects 100 new product groups; 99 contain W3-W6 evaluation events, totaling 52,854 events. Applying the already-frozen P1 rule yields:

Frozen operating point	Coverage	Selective accuracy
0.60	39.16%	76.18%

Frozen operating point	Coverage	Selective accuracy
0.70	21.49%	83.55%
0.80	11.50%	88.14%

The result demonstrates that the development frontier transfers to a much larger non-overlapping set of products without recalibrating P1.

5.3 Untouched representative confirmation

A second representative cohort selects another 100 product groups, again with 99 evaluable groups, and contains 40,474 events. The P1 formula, cutoffs, B0/B1 definitions, time windows, and selection procedure remain frozen.

Frozen operating point	Coverage	Selective accuracy
0.60	41.08%	77.17%
0.70	22.00%	84.78%
0.80	11.01%	90.06%

B1 accuracy on all events in this cohort is 59.37%. At the 0.60 point, the gate therefore raises accuracy by 17.80 percentage points while retaining 41.08% of evaluable events. The important feature is not only the level of accuracy but the close replication of both coverage and accuracy relative to the first new-product batch.

Breadth checks indicate that the pooled result is not driven by only a handful of products. In the confirmatory cohort, 97 products are active at the 0.60 operating point and 93 meet the target rule; at 0.70, 77 are active and 74 meet it; at 0.80, 34 are active and 32 meet it.

5.4 Department challenge

A deliberately broader department-challenge cohort contains 99 evaluable products and 36,027 events. At the frozen 0.60 cutoff:

- B1 accuracy on all events: **61.18%**;
- selective B1 accuracy: **79.33%**;
- event coverage: **40.22%**;
- selective gain: **18.15 percentage points**.

Seven departments contain at least five evaluated products. Six meet the predeclared criterion of at least a ten-point selective gain. Produce improves from 80.19% to 88.81%, a gain of 8.62 points; it is directionally positive but does not satisfy the strict threshold.

This challenge strengthens breadth within the same panel ecosystem. It does not convert the result into evidence of cross-retailer transport.

5.5 Coverage-accuracy frontier

Across both new-product cohorts, stricter P1 thresholds reduce coverage and increase selective accuracy. The first and second batches produce notably similar frontiers despite non-overlapping product selections. This reproducibility is the basis for the evidence grade **CONFIRMED_WITHIN_PANEL_CROSS_PRODUCT**.

From an operational perspective, no single point is universally optimal. The appropriate cutoff depends on the value of coverage relative to prediction error. The study therefore reports the frontier rather than naming one cutoff as globally “best.”

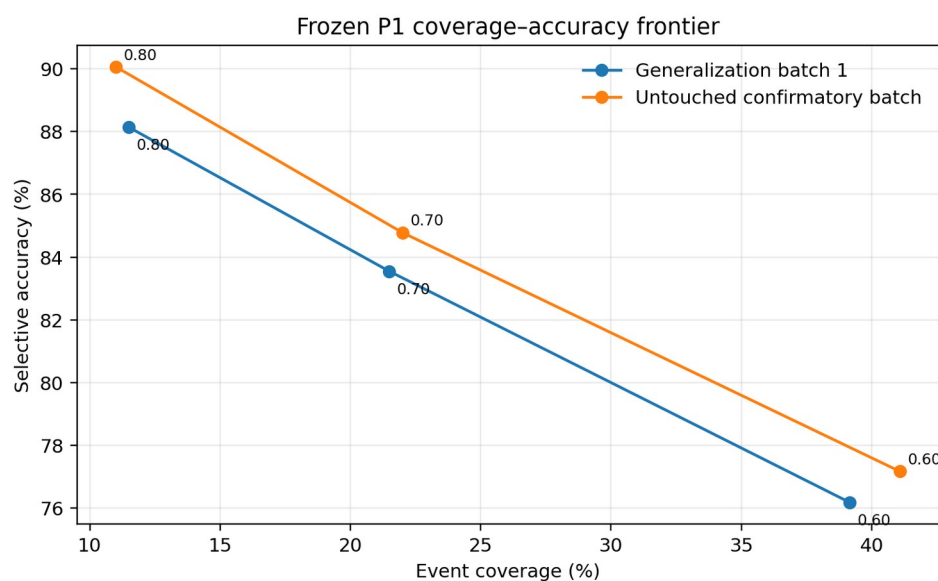


Figure 1. Frozen P1 coverage-accuracy frontier in the two non-overlapping new-product cohorts.

6. Robustness, negative results, and interpretation

6.1 Why the final system is simple

The final B1 + P1 architecture emerged after several more complex paths failed to produce a stable practical improvement. Preserved negative results include learned routers, switch specialists, cold-start rescue, hard-case rescue, short-window product $\times$ model selection, and product-only transforms. Some approaches showed local gains; others harmed performance or failed robustness checks. None provided enough stable evidence to replace the frozen simple architecture.

The project also identified and repaired multiple semantic or engineering defects during development: unreliable row-order state when timestamp semantics were unclear, mixed-category choice sets, cold-start contamination, incomplete ablation, an off-by-one history definition, a control-flag leakage path, and incomplete provenance packaging for an early generalization attempt. A result was not promoted to the final evidence layer merely because it scored well; it had to survive temporal, label, choice-set, leakage, and provenance checks.

This failure history matters because selective-prediction results can otherwise be inflated by post-outcome routing, unstable subgroup definitions, or repeated threshold tuning. The final confirmatory batch therefore prohibits P1 recalibration and learned gates.

6.2 What P1 means — and what it does not mean

P1 is best interpreted as an empirical **pre-event stability score**. High values occur when a household has more historical observations, a more concentrated choice distribution, lower entropy, lower switching, and concentrated recent choices. These features are intuitively compatible with stable choice behavior, but the score does not identify the mechanism that generated that stability.

A high P1 does not imply that the household is globally predictable. Predictability is evaluated for a household $\times$ product-group purchase event. The same household may be predictable in one product group and unpredictable in another. Conversely, a low P1 does not establish irreducible randomness; it indicates that the specific history summary used here does not provide a sufficiently stable signal under the frozen gate.

6.3 Statistical and practical interpretation

The study deliberately separates statistical detectability from practical complexity. B2's incremental gain over B1 is positive under two clustered bootstrap schemes, but the deployment rule requires a larger improvement before accepting extra model complexity. This distinction matters in operational analytics: a statistically nonzero effect can still be too small to justify additional feature pipelines, model maintenance, monitoring, and explanation burden.

The 3.0-point threshold is not offered as a universal standard. It is a predeclared decision rule used to prevent retrospective complexity escalation in this project.

7. Reproducibility and external replication

The public release is designed around reproducibility without redistributing unresolved raw transaction files. It contains the frozen model code and configuration, machine-readable aggregate results, source and output hashes, an evidence ledger, synthetic fixtures, and a strict external-replication runner.

The external runner begins from canonical purchase events and a pre-frozen six-SKU group manifest. The partner does **not** precompute B1 predictions or P1 features. The runner reconstructs market and household histories in chronological order, selects B1 smoothing only within the frozen FIT/TUNE structure, computes P1, applies the frozen cutoffs, and emits aggregate evaluation statistics. Duplicate events, chronology inversions, unknown groups, and choices outside the declared six-SKU set fail closed.

The first external study is intended to be **zero-shot**. The 53-week study window, product-group manifest, and protocol should be frozen before outcome inspection. A successful program run only establishes protocol execution; it does not establish scientific replication. Replication requires an independent environment, adequate sample support, and performance consistent with the predeclared scientific evaluation rule.

This distinction is central to the current evidence grade. New products and departments inside one data ecosystem provide a meaningful confirmation test, but they do not substitute for an independent retailer or consumer panel.

8. Limitations

Five limitations should govern interpretation.

First, the task is conditional six-SKU choice after a purchase occurs. It does not predict whether the household will make a purchase, when the next trip occurs, or what the full next basket contains.

Second, the observed six-product groups are substitute proxies. They do not represent every product available to the consumer at purchase time, and they are not a structural estimate of the true consideration set.

Third, the strongest evidence is within one Complete Journey data ecosystem. The later cohorts are non-overlapping in products and include a department challenge, but the households, data collection process, retail context, and upstream measurement system are not independent across cohorts. The study therefore does not claim external validity to other retailers, countries, panels, or periods.

Fourth, P1 is predictive rather than causal. The data do not establish whether higher predictability is caused by loyalty, habit, stable tastes, lower promotion sensitivity, fewer stock-outs, or other mechanisms. Price elasticity and promotion response are outside the frozen claim set.

Fifth, the project involved substantial exploratory development before the final frozen batches. The strongest claims therefore rest on the later non-overlapping generalization and untouched confirmatory cohorts, not on development-set performance alone.

9. Future research and practical applications

The current results point to a clear development path. MRSP identifies when household history is strong enough to support a high-confidence conditional SKU prediction, and that selective-prediction pattern reproduced across non-overlapping product cohorts. The next stage is to test how far the same signal travels: first across independent retail data, then into broader purchase decisions, response to price and promotion, and real decision workflows. Each step can be tested while keeping the current B1/P1 system frozen as the reference point.

9.1 Independent external replication

The highest-value next experiment is a frozen zero-shot test in an independent retailer or consumer panel. B1, P1, the six-SKU task, cutoffs, abstention rule, and evaluation metrics can be carried over unchanged. If the coverage-accuracy frontier reproduces, the evidence moves from within-panel cross-product confirmation toward cross-environment transportability. If performance changes materially, the new environment identifies where the current signal needs adaptation.

A clean replication sequence is therefore simple: freeze the current method, test Retailer A, use the result to learn, freeze any revised method, and test again in Retailer B or C. This creates a cumulative evidence ladder rather than repeatedly optimizing on one environment.

9.2 From conditional choice to a two-stage purchase system

The current task predicts which SKU will be chosen once a purchase within the six-SKU group occurs. A natural next-generation system is two-stage:

1. first estimate whether a relevant purchase is likely to occur in the coming decision window;
2. then, conditional on purchase, estimate which SKU will be selected and whether the SKU prediction is confident enough to use.

A third decision layer can connect the prediction to action value. This would turn MRSP from a conditional-choice module into a broader purchase-decision system: purchase incidence -> SKU choice -> confidence -> action. The key empirical question is whether P1-like history stability continues to separate predictable from unpredictable events once purchase incidence is introduced.

9.3 Predictability, promotion response, and price sensitivity

The P1 pattern also creates a direct research hypothesis about marketing response. High-P1 histories may reflect stronger preference persistence, loyalty, or inertia, which in turn may be associated with different responses to price changes and promotions. The existing results already show that recent-behavior features add only modest average predictive value beyond household history, making this interaction a natural next test.

A holdout study can estimate promotion lift, switching rates, and price sensitivity across P1 strata. If high-P1 events respond differently from low-P1 events, predictability itself becomes useful for deciding where promotion, retention, or price interventions are most likely to change behavior. This would connect MRSP from prediction quality to marketing allocation.

9.4 Selective deployment and decision value

MRSP's most immediate operational value may be selective deployment rather than universal personalization. P1 already provides a practical routing signal: use the choice model when the event is sufficiently predictable, and abstain when it is not. That logic can sit in front of recommendation ranking, CRM prioritization, campaign targeting, analyst review, or other systems in which errors have different costs.

The production question is therefore a decision frontier: how much coverage, accuracy, intervention cost, and expected value should be traded for a given use case? Low-cost ranking may favor broader coverage; costly

interventions may favor stricter thresholds. A partner pilot can measure this frontier directly and identify the operating point that maximizes decision value rather than raw prediction accuracy.

9.5 Commercialization pathway

If independent replication holds, MRSP has several practical commercialization paths. The first is an evidence and diagnostic service: apply the frozen protocol to a partner's historical data and identify where household-level choices are reliably predictable. The second is a partner pilot that connects selective prediction to an existing recommendation, CRM, promotion, or analytics workflow. The third is a reusable software layer that standardizes scoring, abstention, monitoring, and audit inside the partner's own data environment.

A compact development path is:

Stage	Primary objective	What success enables
Public evidence	Make the frozen MRSP result inspectable and reproducible	Credible external discussion and collaboration
External replication	Test frozen B1/P1 in an independent data environment	Evidence of cross-environment transportability
Partner pilot	Connect selective prediction to a real decision workflow	Measured decision value and an operating point
Paid integration	Adapt MRSP to partner data and production processes	Repeatable delivery and implementation knowledge
Productization	Standardize ingestion, scoring, abstention, monitoring, and audit	A scalable confidence-aware analytics product

The logic is cumulative. Public evidence creates trust; external replication establishes transportability; a partner pilot measures decision value; repeated integrations reveal what can be standardized; productization becomes attractive once the same workflow works across multiple environments.

The current external runner is well suited to this path because it can execute inside a partner's environment and return aggregate metrics and diagnostics. That makes it possible to test MRSP on proprietary retail data without requiring raw transaction records to leave the partner's system.

10. Discussion and conclusion

MRSP's central result is straightforward: household purchase history is a strong predictor of conditional SKU choice, and a pre-event stability score can identify the subset of events on which that prediction becomes much more accurate. The resulting coverage-accuracy frontier reproduced across two non-overlapping new-product cohorts, with the untouched representative batch reaching 77.17% accuracy at 41.08% coverage, 84.78% at 22.00%, and 90.06% at 11.01%.

This changes the modeling question from 'How complex should the predictor become?' to 'When is the current predictor reliable enough to use?' B1 supplies the SKU prediction; P1 determines whether the system answers or abstains. That separation keeps the architecture interpretable while creating a direct control over the accuracy-coverage trade-off.

The development history also points toward simplicity. B2's recent-behavior features add measurable signal, but the gain remains below the frozen practical materiality threshold, and more complex routing strategies did not produce a stable upgrade. The strongest current system is therefore not the most complicated one; it is the simplest model paired with a useful predictability filter.

The present evidence grade is CONFIRMED_WITHIN_PANEL_CROSS_PRODUCT. The natural next upgrade is independent external replication with the frozen protocol. A successful replication would support transportability of the predictability signal across data environments and create a strong basis for partner pilots.

From there, the research program can expand along three connected axes: purchase incidence, price/promotion response, and decision-aware deployment. The first asks whether the signal survives a broader purchase task; the second asks whether predictability also segments behavioral response; the third asks whether selective prediction improves real operational decisions. Together they form a coherent path from empirical regularity to applied decision system.

The commercial opportunity follows the same sequence. MRSP can begin as a transparent diagnostic and validation framework, develop into partner-specific pilots and integration work, and ultimately become a confidence-aware analytics layer if repeated external tests show stable value. The next milestone is therefore concrete: secure one independent data environment, run the frozen replication, and use that result to determine which application path deserves the next investment of effort.

Version note. V2.3 is a publication-metadata update to V2.2, adding the reserved DOI and repository link; the scientific results and claims are unchanged.

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Appendix A. Frozen evidence traceability

This appendix preserves the research-audit boundary behind the public prose. It is not a substitute for the machine evidence package.

Claim area	Frozen evidence
Task definition	Conditional six-alternative SKU choice; no purchase-incidence or next-basket claim
Development baseline	17,408 events; B0 37.28%, B1 64.53%, B2 65.52%
Practical B1/B2 decision	26,659 nonzero-history events; +1.2604 pp B2-B1; 3.0 pp materiality threshold; retain B1
P1	Parameter-free pre-event stability score; no current label; frozen formula and cutoffs
Generalization batch 1	99 evaluable products; 52,854 events; frozen frontier reproduced
Untouched representative batch	99 evaluable products; 40,474 events; 41.08%/77.17%, 22.00%/84.78%, 11.01%/90.06% coverage/accuracy
Department challenge	99 evaluable products; 36,027 events; 40.22% coverage, 79.33% selective accuracy; six of seven supported departments meet the strict gain rule
Evidence grade	CONFIRMED_WITHIN_PANEL_CROSS_PRODUC T; external retailer replication pending

The authoritative scientific source artifact is bound by SHA-256

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